RENSEE GAJIPARA

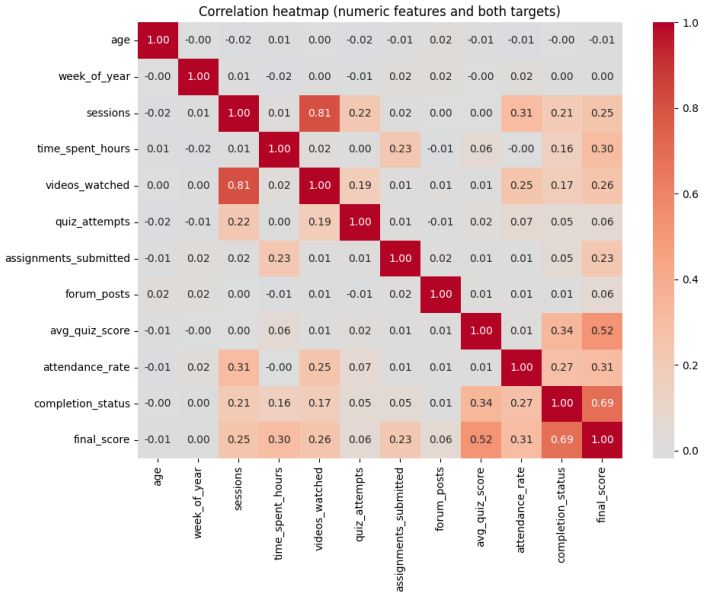
Final Report - Smart Outcome Predictor

Ensemble-based prediction of course completion and final score

1. Objective and Dataset

An EdTech platform wants to (1) predict whether a student will complete a course (classification) and (2) predict the final score (regression), so it can support at-risk students early.

- 5,200 students, 17 features after dropping student_id and course_start_date; 5 categorical and 10 numeric features.
- Only 37.5% of students completed the course (class imbalance), so an 80/20 stratified split was used (random_state=42).
- Missing values: time_spent_hours (112), avg_quiz_score (81), attendance_rate (80).
- avg_quiz_score is the strongest predictor (correlation +0.34 with completion, +0.52 with final score). Each target was excluded from the other task's features to avoid leakage.



2. Methodology

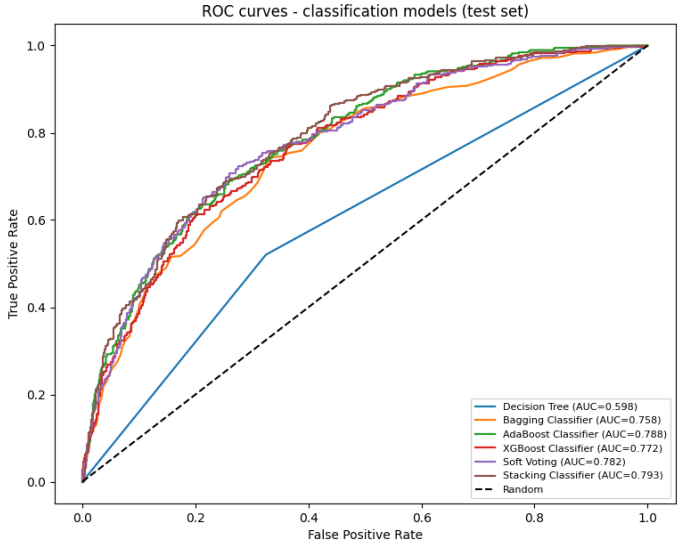
- **Preprocessing pipeline:** numeric = StandardScaler + KNNImputer; categorical = constant fill + OneHotEncoder (30 features), fitted on training data only.
- **Models:** Decision Tree (baseline), Bagging (100 trees), AdaBoost, XGBoost, Hard Voting, Soft Voting, Stacking (LR, KNN, DT, SVM base models; LR/Ridge meta-learner, 5-fold CV).
- **Metrics:** Accuracy, Precision, Recall, F1, ROC-AUC (classification); MAE, RMSE, R^2 (regression).
- **Deployment:** best models saved as full pipelines in pickle files and used by a Streamlit app.

3. Results

3.1 Classification (test set)

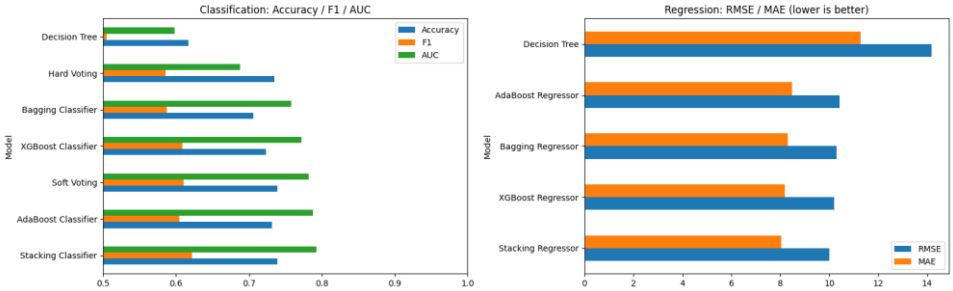
Model	Accuracy	Precision	Recall	F1	AUC
Stacking Classifier	0.739	0.682	0.572	0.622	0.793
AdaBoost Classifier	0.732	0.676	0.546	0.604	0.788
Soft Voting	0.739	0.695	0.544	0.610	0.782
XGBoost Classifier	0.723	0.647	0.574	0.609	0.772
Bagging Classifier	0.706	0.619	0.559	0.588	0.758
Hard Voting	0.735	0.707	0.500	0.586	0.688
Decision Tree	0.617	0.490	0.520	0.505	0.598

Results are also saved in Outputs/classification_results.csv and Outputs/regression_results.csv.



3.2 Regression (test set)

Model	MAE	RMSE	R^2
Stacking Regressor	8.03	9.99	0.451
XGBoost Regressor	8.20	10.19	0.429
Bagging Regressor	8.31	10.30	0.417
AdaBoost Regressor	8.49	10.41	0.405
Decision Tree	11.28	14.19	-0.105



4. Comparison and Analysis

Bagging vs Boosting. Bagging reduced the Decision Tree's variance (AUC 0.598 to 0.758; RMSE 14.19 to 10.30) but its trees still fit the training data perfectly. Boosting generalised slightly better (AdaBoost AUC 0.788; XGBoost RMSE 10.19) by correcting errors sequentially.

AdaBoost vs XGBoost. AdaBoost is simple and strong for classification but is the weakest ensemble regressor. XGBoost is the best boosting regressor and has better recall thanks to regularisation, shrinkage and subsampling; it overfits more if left unconstrained (train accuracy 0.835 vs test 0.723).

Voting vs Stacking. Soft voting beat hard voting (AUC 0.782 vs 0.688) because it keeps model confidence. Stacking was best on both tasks because its meta-learner weights diverse base models, at the cost of longer training and more complex deployment.

Single tree vs ensembles. Every ensemble clearly beat the single Decision Tree on all metrics.

5. Final Recommendation

- **Deploy:** Stacking Classifier (completion) and Stacking Regressor (final score) as the saved pipelines completion_pipeline.pkl and score_pipeline.pkl.
- **Alternative:** XGBoost is a close and cheaper option (AUC 0.772, RMSE 10.19) if speed or simplicity matters.
- **Avoid:** a single Decision Tree.

6. Business Interpretation

- The platform can flag students with low completion probability (accuracy 74%, precision 0.68) for reminders, mentoring or quiz support.
- Final scores can be forecast within about 8 points on average (MAE 8.0).
- Quiz performance, attendance, sessions and video engagement are the strongest drivers, so interventions should focus on them.
- Recall is only 0.57, so the tool should support human review, not replace it. Richer data (e.g. weekly progress trends) is the best route to improvement.

